

BUSINESS UNDERSTANDING REPORT

Week 1: Machine Learning Problem Framing & Data Understanding

Project Title: Customer Churn Prediction for ABC Communications Ltd

INTRODUCTION:

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from historical data and make predictions or decisions without being explicitly programmed for every situation. Organizations use machine learning to automate processes, improve decision-making, and gain valuable insights from data.

Predictive Analytics is the process of using historical data, statistical techniques, and machine learning algorithms to predict future outcomes. Businesses use predictive analytics to forecast customer behavior, sales, demand, and risks.

Customer Churn refers to the situation where customers stop using a company's products or services. High customer churn reduces revenue, increases customer acquisition costs, and negatively affects business growth. Predicting churn allows businesses to identify customers who are likely to leave and take proactive measures to retain them.

Business Problem

ABC Communications Ltd is experiencing customer churn, which results in revenue loss and reduced customer retention. The company needs a machine learning solution that can identify customers who are likely to leave before they actually cancel their services.

By identifying customers at risk of churning, the company can implement targeted retention strategies such as discounts, personalized offers, improved customer support, or loyalty programs to reduce customer loss.

Business Objective

The objective of this project is to develop a machine learning model that predicts whether a customer is likely to churn based on customer information and service usage.

The prediction will help the company:

- Improve customer retention.
- Reduce revenue loss.
- Support better business decisions.
- Increase customer satisfaction through proactive intervention.

Machine Learning Objective

The machine learning objective is to build a predictive model that classifies customers into one of two categories:

- Customers who are likely to churn.
- Customers who are likely to remain with the company.

The model will learn patterns from historical customer data and use these patterns to predict future customer behavior.

Business Questions and Answers:

- Question 1: What business problem should the model solve?

The model should predict customer churn so that ABC Communications Ltd can identify customers who are likely to leave and take appropriate actions to retain them before they cancel their services.

- Question 2: What is the target variable?

The target variable is "Churn". This variable indicates whether a customer has left the company or continues using its services.

From the dataset, there are:

- 5,174 customers who did not churn
- 1,869 customers who churned

- Question 3: Which input features should be used?

The dataset contains 21 columns. The "customerID" column should not be used as an input feature because it is an identifier for individual customers rather than a meaningful customer characteristic.

The potential input features are:

"gender", "SeniorCitizen", "Partner", "Dependents", "tenure", "PhoneService", "MultipleLines", "InternetService", "OnlineSecurity", "OnlineBackup", "DeviceProtection", "TechSupport", "StreamingTV", "StreamingMovies", "Contract", "PaperlessBilling", "PaymentMethod", "MonthlyCharges", "TotalCharges"

These features contain information about customers' demographics, services, contracts, tenure, and billing.

➤ Question 4: Is this classification or regression?

This is a supervised machine learning classification problem, specifically a binary classification problem. The reason is that the target variable "Churn" has only two possible outcomes: "Yes" and "No".

The model will therefore be designed to classify customers into one of these two categories.

➤ Question 5: Which evaluation metrics should be used?

Suitable evaluation metrics include:

- Accuracy — measures the overall proportion of correct predictions.
- Precision — measures how many customers predicted to churn actually churn.
- Recall — measures how many of the actual churners are correctly identified.
- F1-score — provides a balance between precision and recall.
- ROC-AUC — measures the model's ability to distinguish between churned and non-churned.
- Confusion Matrix

Recall and F1-score should receive particular attention because the dataset contains more customers who did not churn than customers who churned. Relying only on accuracy could therefore give an incomplete picture of model performance.

➤ Question 6: Which machine learning algorithms are suitable?

Several supervised machine learning algorithms are suitable for this classification task, including:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naïve Bayes
- Gradient Boosting
- XGBoost

These algorithms can learn patterns from historical customer data and classify whether a customer is likely to churn.

Expected Business Impact

A successful churn prediction model can provide several benefits to ABC Communications Ltd:

- Higher customer retention.
- Reduced customer acquisition costs.
- Increased profitability.
- Better marketing strategies.
- Improved customer satisfaction.
- Data-driven decision-making.

Conclusion

The customer churn problem at ABC Communications Ltd can be framed as a binary classification problem in which "Churn" is the target variable and customer demographic, service, contract, tenure, and billing information are potential input features.

The dataset will first be understood and explored to identify relevant patterns and data-quality issues. The proposed algorithms are Logistic Regression, Decision Tree, and Random Forest, while Accuracy, Precision, Recall, F1-score, and ROC-AUC are proposed as evaluation metrics.

This business understanding provides the foundation for the subsequent machine-learning development stages.